

INTELLIGENT REAL-TIME SURVEILLANCE USING YOLOV8 FOR WEAPON DETECTION AND DEEPFACE FOR CRIMINAL IDENTIFICATION

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Abstract - Rising security concerns and criminal incidents in public environments have driven demand for intelligent, automated surveillance capable of identifying threats in real time. Conventional CCTV-based monitoring depends largely on human operators, introducing limitations such as delayed response times, operator fatigue, and inconsistent vigilance. This paper presents an AI-driven surveillance framework that combines deep learning methodologies for automated weapon detection and the identification of individuals with criminal records. The system captures live video from CCTV or USB camera feeds and continuously processes incoming frames using the YOLOv8 object detection architecture to identify potentially dangerous weapons. Simultaneously, facial recognition is performed through the DeepFace framework, which compares detected faces against a pre-established database of known offenders. A dedicated threat evaluation module consolidates outputs from both detection components and triggers alerts upon identifying suspicious behavior. Security personnel can monitor live

detection results and notifications through an integrated web-based dashboard. The proposed system offers a meaningful contribution toward improving public safety.

Keywords:

Real-Time Surveillance, Weapon Detection, Facial Recognition, YOLOv8, Deep Learning.

I. INTRODUCTION

The importance of public safety in an urban environment has become a critical concern, given the increasing number of violent incidents and armed threats in a shared public space. In general, the use of traditional CCTV systems, although prevalent in airports, transport hubs, and business environments, remains a passive system that relies entirely on the attention of a human observer. In reality, this condition cannot be met as the observer may become distracted, have a poor understanding of the situation, or may not have the capacity to process all the information being received. In this regard, security-critical situations involving weapons or individuals who have a prior criminal record often go unnoticed until the situation arises, greatly reducing

the effectiveness of the traditional security system.

With the recent breakthroughs in Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL), there is now a realistic way forward to develop intelligent and autonomous surveillance systems. Single-stage object detection models like YOLO have reached the milestone of real-time processing with high accuracy, and deep learning-based facial recognition systems can now provide accurate identity verification even in challenging imaging scenarios like occlusion, low resolution, and lighting variations. In this paper, we introduce IntellEye, an AI-assisted surveillance system that combines YOLOv8-based weapon detection and DeepFace-based facial recognition into a single web-based surveillance platform. The proposed system can automatically analyze live video streams from CCTV cameras or USB cameras, simultaneously detect weapons and identify criminals, and provide instant notifications through a centralized dashboard.

II. PROBLEM STATEMENT

The existing infrastructure for public surveillance is ineffective in addressing the issue of weapon-related occurrences due to its very nature, which requires constant human intervention. The majority of existing CCTV systems are only used for passive surveillance, whereby video recordings are made without any form of analysis. The humans assigned to oversee these systems are prone to fatigue, as they are required to process multiple video feeds at once over an extended period. This makes it impossible to identify armed threats or known offenders, as the situation is likely to get out of hand before any action can be taken.

In addition, existing automated surveillance systems are limited, as they only focus on specific areas, such as weapon detection or identity verification, but not

both. The practical environment, such as lighting, obstacles, and high concentrations of people, affects the overall accuracy and reliability of such systems. These limitations combine to demonstrate the lack of a comprehensive and intelligent surveillance system that is able to simultaneously identify weapons, recognize known criminals, and send instant alerts to security personnel via a user-friendly and centralized monitoring interface.

III. CHRONOLOGICAL REVIEW OF LITERATURE

Weapon Detection using Deep Learning

Yeddula and Reddy (2022) carried out a research work on the automated detection of weapons in CCTV footage by incorporating a YOLOv5 deep learning model into the surveillance system. Based on convolutional feature extraction, trained on annotated weapon images, the model was tested on real-world CCTV footage to detect weapons and sharp objects. Parameters like detection confidence, bounding box precision, and latency were taken into consideration. From the findings, it was inferred that deep learning-based detection is far more superior to traditional image processing techniques and offers a sound platform for real-time threat identification in public surveillance systems.

Dass et al. (2024) developed a real-time weapon detection system using YOLOv7-Tiny for implementation in public surveillance systems. The research work focused on the compromise between model complexity and detection accuracy by using a lightweight version of the YOLO model, which is more suitable for edge computing. The proposed system was tested in various surveillance settings and concluded that YOLOv7-Tiny is a competitive detection model with much faster inference times than complex models.

Bushra et al. (2022) analyzed the use of YOLOv5 for weapon detection in densely

populated and crowded public settings. The authors analyzed the difficulties caused by occlusion, lighting, and overlapping objects in real-world surveillance videos. A YOLOv5 model was trained and tested on crowd-sourced video datasets, proving that single-stage detectors preserve accurate identification performance even in visually complex settings. It was concluded that smart video surveillance systems fueled by optimized YOLO models provide a scalable solution for proactive public safety monitoring.

Raj et al. (2024) introduced a hybrid deep learning model that combines convolutional neural network-based detection with K-Means clustering for weapon localization in CCTV surveillance systems. The new model first employs clustering to identify regions of interest in video frames before submitting potential regions to the detection model, thus suppressing false positives and improving localization accuracy. The new model was tested on benchmark surveillance datasets and proved to provide better detection accuracy than pure deep learning models, indicating that the combination of unsupervised scene segmentation and supervised detection improves system performance.

Criminal Activity Recognition and Face Identification

Arpa et al. (2023) developed an integrated model that utilized machine learning and deep learning approaches simultaneously for criminal activity recognition and face identification from surveillance videos. The proposed model consisted of a two-branch framework, where one branch was responsible for activity recognition based on spatiotemporal features, and the other branch was responsible for face identification based on embedding comparison for facial verification. The experimental results on publicly available crime and face recognition datasets showed

enhanced recognition accuracy compared to individual modality-based approaches. The authors concluded that the joint analysis of behavioral and biometric information in a single framework significantly enhances the ability of automated surveillance systems to detect known criminals in real-time.

Veesam and Satish (2025) developed an integrated model that utilized multimodal fusion for video summarization and crime scene analysis from surveillance videos. The proposed framework processed raw video inputs to produce summarized versions that emphasized anomalous activity while simultaneously identifying crime scenes at the scene level. The experimental results on forensic surveillance video datasets showed that multimodal fusion provided enhanced situational awareness and accurate event localization compared to unimodal approaches.

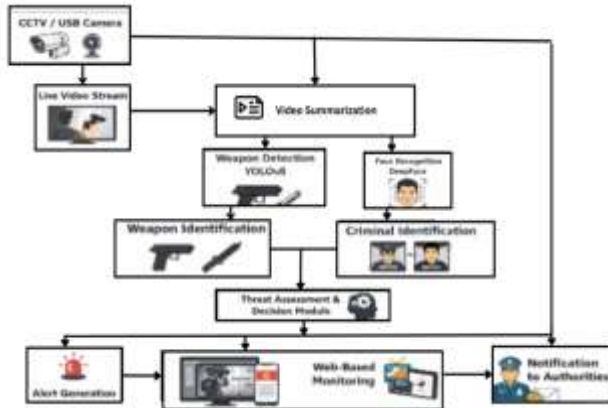
IV. PROPOSED FRAMEWORK

The proposed AI-based Weapon and Criminal Detection System is designed using a modular pipeline architecture that is scalable, real-time, and easy to deploy. The pipeline starts with a Video Acquisition Layer where OpenCV is used to continuously acquire live frames from CCTV or USB cameras. Every frame is then preprocessed with operations such as resizing, pixel-level normalization, and format conversion to make it compatible with the subsequent deep learning models.

- The Weapon Detection Layer uses YOLOv8 to detect weapons in every preprocessed frame. YOLOv8 is a single-stage detector that divides the image into a spatial grid and predicts the bounding box coordinates and class probabilities in a single pass, achieving real-time processing speed with high accuracy. Every detection result contains the class label of the

weapon, the bounding box coordinates, and a confidence score.

- The Face Recognition Layer uses DeepFace to locate faces in each frame, followed by the extraction of high-dimensional facial embedding vectors. These vectors are then compared to pre-existing reference vectors in the offender database using cosine similarity, and those above a specified threshold are determined to be confirmed offender identifications.
- The Threat Assessment Layer is the decision-making engine of the system, using rule-based reasoning on the joint output of the detection and recognition components. A confirmed offender identification increases the threat priority level, and the presence of both weapon detection and offender identification sends a critical priority alert. The Alert and Monitoring Layer notifies



the user of threat events

Fig.1. Architectural Design of AI-Powered Weapon and Criminal Detection Framework

V. PROBLEM FORMULATION AND MATHEMATICAL MODELLING

YOLOv8 computes a detection confidence score for each candidate weapon class w in frame F as:

$$Conf(w, F) = P(object) \times IoU(predicted, ground-truth) \quad (1)$$

where $P(object)$ represents the objectness probability and IoU denotes the Intersection over Union between the predicted and ground-truth bounding boxes. A detection is accepted only when $Conf(w, F)$ exceeds the weapon threshold τ_w .

DeepFace computes the cosine similarity between a query embedding E_q and a stored database embedding E_d as:

$$CosineSimilarity(E_q, E_d) = (E_q \cdot E_d) / (||E_q|| ||E_d||) \quad (2)$$

where $E_q \cdot E_d$ is the dot product of the two vectors and $||E||$ denotes the L2 norm. An identity match is confirmed when this similarity exceeds threshold τ_f .

The combined threat level T for a given frame is defined as:

$$T = \alpha \times W_{detect} + \beta \times F_{match} \quad (3)$$

where $W_{detect} \in \{0,1\}$ indicates whether a weapon has been detected, $F_{match} \in \{0,1\}$ indicates whether a criminal face match has been confirmed, and α, β are weighting coefficients. When T surpasses a critical threshold, a high-priority alert is dispatched. Fig.2 shows the YOLOv8 training loss curve across 50 epochs.

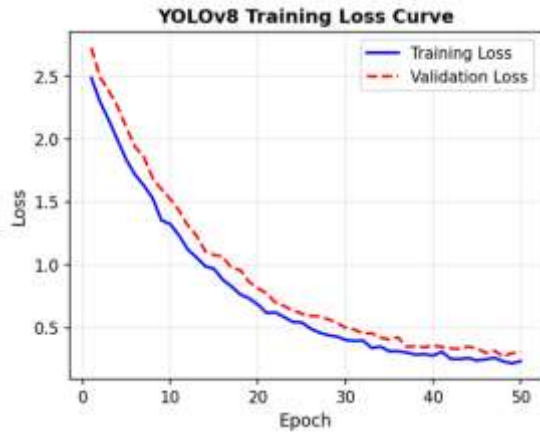


Fig.2. YOLOv8 Training and Validation Loss Curve

VI. RESULTS AND DISCUSSION

The system was assessed using a dataset of recorded surveillance footage and live test video streams containing weapon instances and subjects with known facial identities.

Parameter	Value
Detection Model	YOLOv8
Face Recognition	DeepFace
Weapon Classes	Gun, Knife
Web Interface	Flask (Python)
Processing Mode	Real-Time
Input Source	CCTV / USB Camera

Table.1. System Configuration Summary

Table 2 presents the weapon detection performance of the YOLOv8 model, reporting precision, recall, and F1-score for individual weapon classes including gun and knife, along with overall detection performance across the dataset.

Weapon Class	Precision	Recall	F1-Score
Gun	0.92	0.89	0.90

Knife	0.88	0.85	0.86
Overall	0.90	0.87	0.88

Table.2. YOLOv8 Weapon Detection Performance

Table 3 illustrates the criminal recognition performance of the DeepFace framework evaluated across three imaging conditions, namely standard, low-light, and partial occlusion, measuring accuracy, false positive rate, and recognition speed.

Metric	Standard	Low Light	Partial Occlusion
Accuracy (%)	94.2	81.3	78.6
False Positive Rate	0.04	0.11	0.14
Recognition Speed (ms)	38	41	43

Table.3. DeepFace Criminal Recognition Performance

Table 4 summarises the time complexity of each component within the IntelliEye pipeline, covering YOLOv8 weapon detection, DeepFace embedding extraction, database similarity search, and threat assessment processing.

Component	Time Complexity
YOLOv8 Weapon Detection	$O(F \times n)$
DeepFace Embedding Extraction	$O(F \times d)$
Database Similarity Search	$O(d \times k)$
Threat Assessment	$O(1)$

Table.4. Algorithm Time Complexity

Fig.3 shows the confusion matrix of the threat prediction module. The model correctly identifies the majority of true threat and non-threat frames, with a low false negative rate critical for surveillance applications.

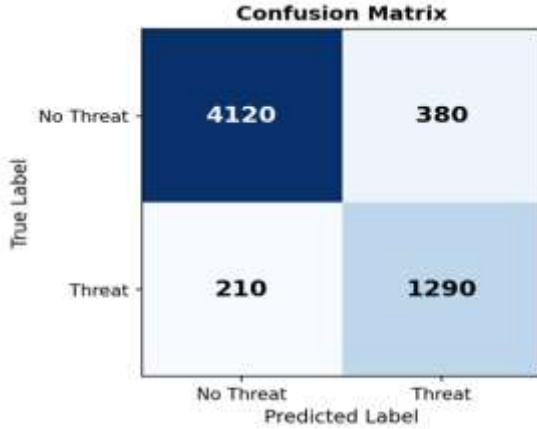


Fig.3. Confusion Matrix of Threat Detection Module

Fig.4 presents the Precision-Recall curves for gun and knife detection classes.

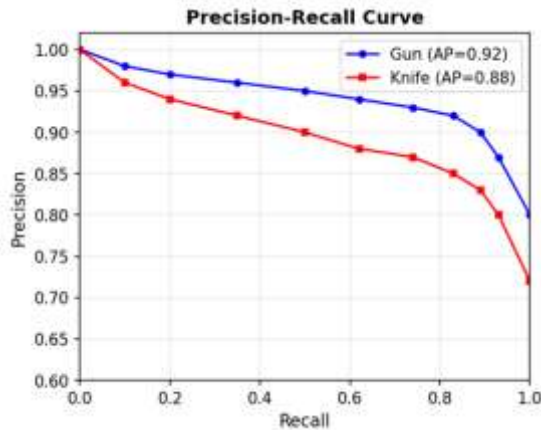


Fig.4. Precision-Recall Curve for Weapon Detection Classes

Fig.5 compares the proposed YOLOv8-based system against existing methods in terms of precision and recall. The proposed framework outperforms

YOLOv5 and YOLOv7-tiny baselines, achieving the highest precision (0.90) and recall (0.87) among all compared methods.

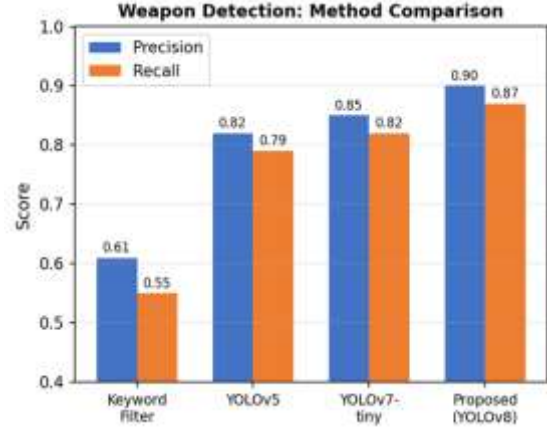


Fig.5. Weapon Detection Performance Comparison Across Methods

Fig.6 illustrates the face recognition accuracy of the DeepFace module across four environmental conditions. Accuracy of 94.2% under standard lighting confirms the robustness of the recognition pipeline, while performance degradation under low-light and partial occlusion conditions highlights areas for future improvement.

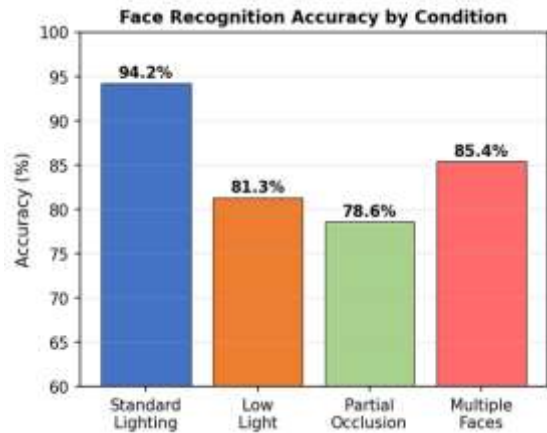


Fig.6. Face Recognition Accuracy Under Different Environmental Conditions

Fig.7 shows the end-to-end processing latency distribution across 500 test frames. The mean processing latency of approximately 40 ms per frame confirms

real-time capability, as the system comfortably maintains performance above the 25 fps threshold required for live surveillance.

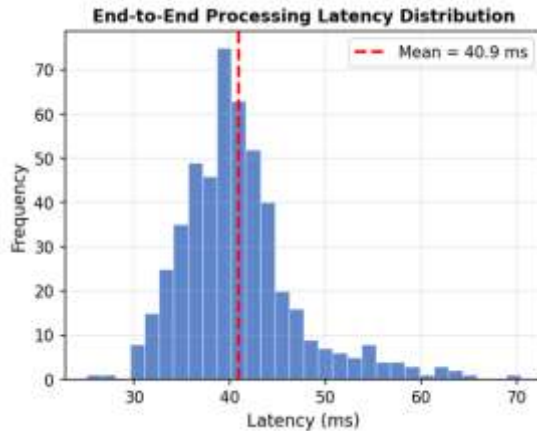


Fig.7. End-to-End Processing Latency Distribution

The results demonstrate the effectiveness of the proposed integrated framework. The weapon detection module achieved an overall F1-score of 0.88, while the face recognition module achieved 94.2% accuracy under standard conditions. The integration of both modules into a unified threat assessment pipeline provided enhanced detection intelligence compared to standalone systems, enabling multi-level threat classification that single-task systems cannot provide.

VII. AVENUES FOR FUTURE RESEARCH

- It is proposed that the IntelliEye framework should be enhanced to include night vision and infrared camera capabilities, allowing for the detection of weapons and identification of criminals in situations involving low visibility and zero-visibility surveillance.
- It is proposed that additional research should be conducted to

enhance the IntelliEye framework to include multi-camera synchronisation, allowing for the tracking and identification of criminals in situations involving multiple cameras in public spaces.

- It is proposed that the facial recognition module should be enhanced to include multi-angle and partial face detection capabilities, allowing for increased accuracy in identifying criminals.

VIII. CONCLUSION

This paper proposed an integrated AI-powered surveillance framework combining YOLOv8-based weapon detection and DeepFace-based criminal face recognition for real-time threat monitoring. The system demonstrated strong detection performance with an overall weapon detection F1-score of 0.88 and face recognition accuracy of 94.2% under standard conditions. The Flask-based web interface enables accessible real-time monitoring with annotated video feeds and automated alerts.

The modular architecture ensures independent scalability of each component and supports integration into existing CCTV infrastructure. Future research directions include the adoption of transformer-based object detection models, dynamic criminal database updates, multi-camera network integration, edge deployment for resource-constrained environments, and fairness-aware optimization to reduce bias in face recognition systems.

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